

Project Name: LaunchPad AI

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Max Marks: 4 Marks

Solution Requirements

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform. Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do). Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

STEP 1: FUNCTIONAL REQUIREMENTS (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	System protects application boundaries by initializing API routes through localized parameter checks or key validation hooks rather than requiring public portal accounts for immediate utility evaluation.	Medium
2	Authorization levels	Enforces roles separating standard Innovator users (who run business validation flows and view outputs) from platform Admins (who update prompt instructions and examine configuration metadata).	Medium
3	External interfaces	Establish reliable connections to Google Gemini AI API via secure SDK parameters, and route local Flask server traffic to the web using secure Ngrok endpoints.	High
4	Transactions processing	Manages state transactions securely by ensuring structured input text forms map completely to downstream prompt formatting matrices without losing context or causing timeout faults.	High
5	Reporting	Generates and renders dynamic, comprehensive business evaluation dashboards containing viability metrics, structured SWOT grids, and strategic monetization frameworks.	High

S.No	Requirement Category	Requirement Description	Priority
6	Business rules, etc.	Restricts submission blocks to valid character lengths and ensures that prompt engineering logic strictly applies Gemini 1.5 Flash models to ensure structured output generation.	High
7	Compliance to laws	Adheres to data privacy principles by utilizing an environment-based configuration (.env) to avoid exposing individual cloud keys or secret strings publicly.	High
8	Other	Includes responsive UI component state handling, providing immediate visual loading cards or status indicators during live API data fetching intervals.	Medium

STEP 2: NON-FUNCTIONAL REQUIREMENTS (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The web interface must load efficiently, and backend processes must handle response processing quickly.	UI rendering completes in < 2 seconds; AI report parsing maps to dashboard layout immediately after API response stream closes.
2	Scalability	The deployment design must easily accommodate fluctuating user evaluation traffic volumes.	The Flask architecture can support multiple concurrent requests concurrently using standard asynchronous threading parameters.
3	Security & Data Privacy	Protects sensitive credentials and cloud configurations from public disclosure.	100% of local tokens are read from environment configurations (.env); direct keys are hidden from open application logs.
4	Reliability & Availability	Ensures the application service is accessible when the public URL tunnel is active.	Maintains consistent live connectivity with > 99% uptime during active project pitch testing windows.
5	Usability & Accessibility	The interactive dashboard layout must adjust fluidly to different hardware setups.	Fully fluid responsive interface across mobile, tablet, and desktop viewports.
6	Other (Maintainability)	The codebase should follow structured modular logic for easy framework changes or upgrades.	Clean separation of route architectures, AI generation scripts, and HTML template folders.